



UDAY TITANS

ONE SHOT SERIES

Mathematics

Relations And Functions

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Topics to be Covered



Relations & Functions

Class 11th



Ordered Pairs



An ordered pair consists of two objects or elements in a given fixed order.

→ $(1, 2)$

→ $(2, 1)$

→ (a, b)

→ $(\text{Kartika}, \text{Rashmi})$



An ordered pair is not a set consisting of two elements. The ordering of the two elements in an ordered pair is important and the two elements need not be distinct.



Cartesian Product of Sets



Cartesian Product of Sets:

Let A and B be any two non-empty sets. The set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$ is called the cartesian product of set A and B and is denoted by $A \times B$.

$$A = \{1, 2\}$$

$$B = \{3, 4\}$$

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$B \times A = \{(3, 1), (3, 2), (4, 1), (4, 2)\}$$

$$A \times B \neq B \times A \quad \text{In general}$$



Number of Elements in the Cartesian Product of Two Sets

$$n(A) = 5$$

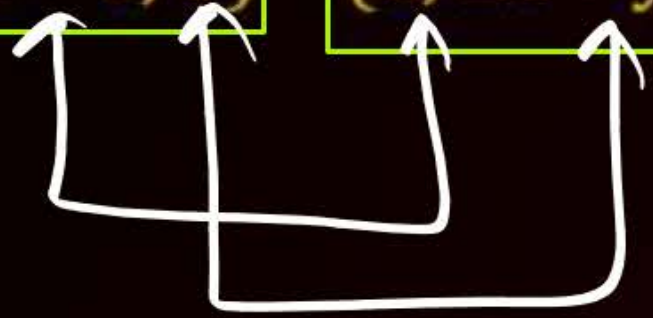
$$n(B) = 3$$

$$n(A \times B) = n(A) \times n(B)$$

QUESTION



Find x and y , if $(x + 3, 5) = (6, 2x + y)$.



$$x + 3 = 6$$

$$x = 3$$

$$5 = 2x + y$$

$$5 = 2(3) + y$$

$$5 = 6 + y$$

$$-1 = y$$

QUESTION



If $A = \{1, 3, 5, 6\}$ and $B = \{2, 4\}$, find $A \times B$ and $B \times A$.

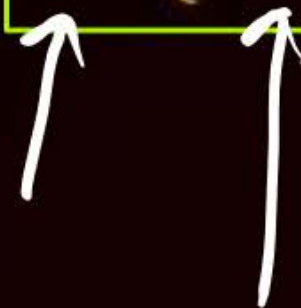
$$A \times B = \{(1, 2), (1, 4), (3, 2), (3, 4), (5, 2), (5, 4), (6, 2), (6, 4)\}$$

$$B \times A = \{(2, 1), (2, 3), (2, 5), (2, 6), (4, 1), (4, 3), (4, 5), (4, 6)\}$$

QUESTION



If $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{1, 3, 5\}$, find $A \times (B \cap C)$



$$A = \{1, 2, 3\}$$

$$B \cap C = \{3\}$$

$$A \times (B \cap C) = \{(1, 3), (2, 3), (3, 3)\}$$

QUESTION



If $a \in \{-1, 2, 3, 4, 5\}$ and $b \in \{0, 3, 6\}$, write the set of all ordered pairs (a, b) such that $a + b = 5$.

$\{(-1, 0), (-1, 3), (-1, 6), (2, 0), (2, 3), (2, 6), (3, 0), (3, 3), (3, 6), (4, 0), (4, 3), (4, 6), (5, 0), (5, 3), (5, 6)\}$

Ans: $\{(-1, 6), (2, 3), (5, 0)\}$

QUESTION



If $a \in \{2, 4, 6, 9\}$ and $b \in \{4, 6, 18, 27\}$, then form the set of all ordered pairs (a, b) such that a divides b and $a < b$.

$$\text{Ans} = \{(2, 4), (2, 6), (2, 18), (6, 18), (9, 18), (9, 27)\}$$

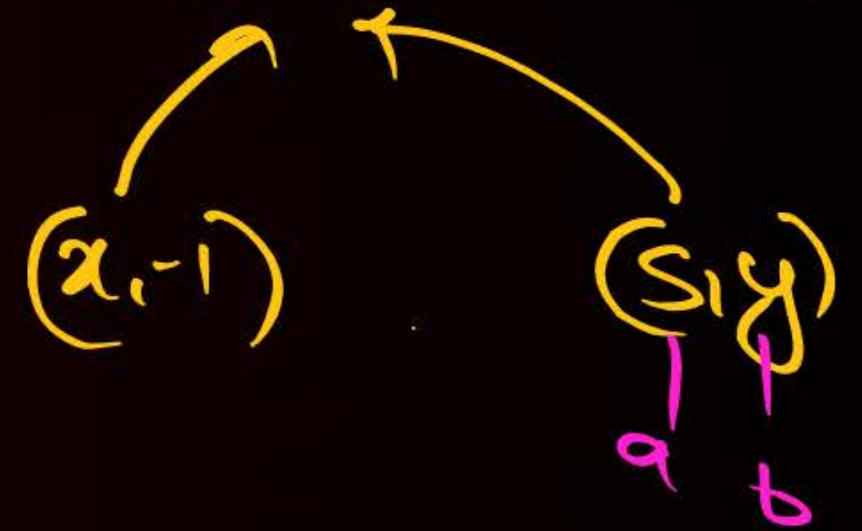
QUESTION



If the ordered pairs $(x, -1)$ and $(5, y)$ belong to the set $\{(a, b) : b = 2a - 3\}$, find the values of x and y

Let A

$$A = \{(a, b) : b = 2a - 3\}$$



$$\begin{aligned} y &= 2(5) - 3 \\ y &= 10 - 3 \\ y &= 7 \end{aligned}$$

$$\begin{aligned} b &= 2a - 3 \\ -1 &= 2(x) - 3 \\ -1 &= 2x - 3 \\ -1 + 3 &= 2x \\ 2 &= 2x \end{aligned}$$

$$1 = x$$

QUESTION



If $A = \{-1, 1\}$, find $A \times A \times A$.

$$A = \{-1, 1\}, A = \{-1, 1\}$$

$$A \times A = \{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$$

$$A = \{-1, 1\}$$

$$A \times A \times A = \{(-1, -1, -1), (-1, 1, -1), (1, -1, -1), (1, 1, -1), (-1, -1, 1), (-1, 1, 1), (1, -1, 1), (1, 1, 1)\}$$

QUESTION

If $A = \{1, 2, 4\}$, $B = \{2, 4, 5\}$, $C = \{2, 5\}$ then $(A - B) \times (B - C)$ is

- A** $\{(1, 2), (1, 5), (2, 5)\}$
- B** $\{(1, 4)\}$
- C** $(1, 4)$
- D** None of these

Ans

QUESTION



State whether each of the following statement are true or false. If the statement is false, re-write the given statement correctly:

- (i) If $P = \{m, n\}$ and $Q = \{n, m\}$, then $P \times Q = \{(m, n), (n, m)\}$ *False.*
- (ii) If A and B are non-empty sets, then $A \times B$ is a non-empty set of ordered pair (x, y) such that $x \in B$ and $y \in A$. *False.*
- (iii) If $A = \{1, 2\}$, $B = \{3, 4\}$, then $A \times (B \cap \phi) = \phi$. *True*

$$(i) \quad n(P \times Q) = 4$$

$$P \times Q = \{(m, n), (m, m), (n, n), (n, m)\}$$

$$(ii) \quad x \in A, y \in B.$$

$$\begin{aligned} (iii) \quad A \times (B \cap \phi) \\ &= A \times \phi \\ &= \phi. \end{aligned}$$

QUESTION

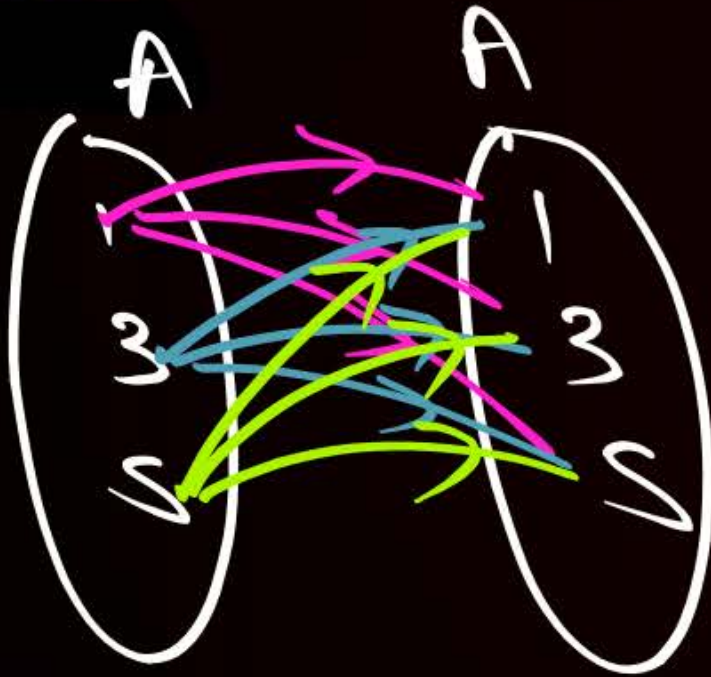


If $A = \{1, 3, 5\}$, $B = \{x, y\}$ represent the following products by arrow diagrams:

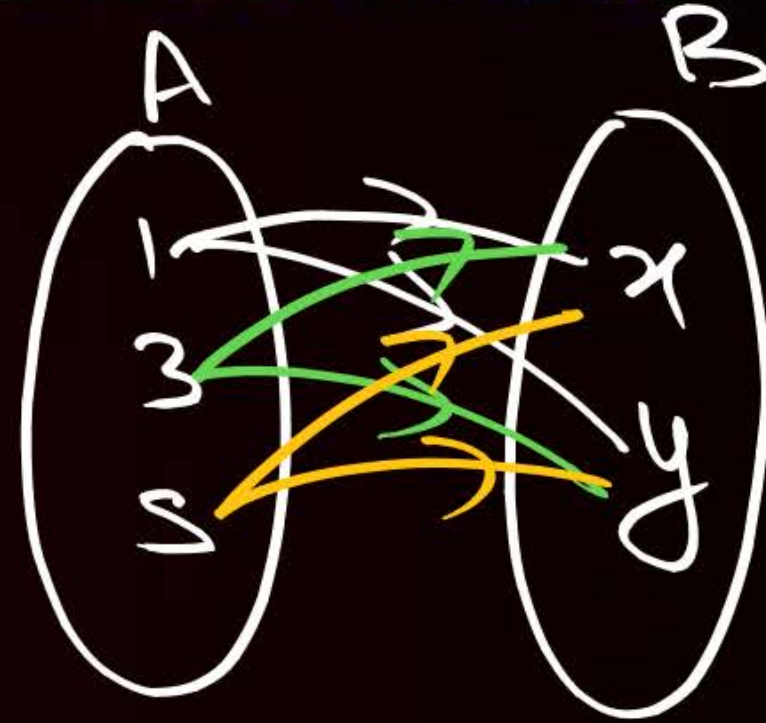
(i) $A \times B$

(ii) $A \times A$

(ii)



(i)



QUESTION



Let A and B be two sets such that $A \times B$ consists of 6 elements. If three elements of $A \times B$ are: $(1, 4), (2, 6), (3, 6)$. Find $A \times B$ and $B \times A$. How

$$n(A \times B) = 6$$

$$A \times B = \{(1, 4), (2, 6), (3, 6), (1, 6), (2, 4), (3, 4)\}$$

$$\checkmark A = \{1, 2, 3\}$$

$$\checkmark B = \{4, 6\}$$



Ex:

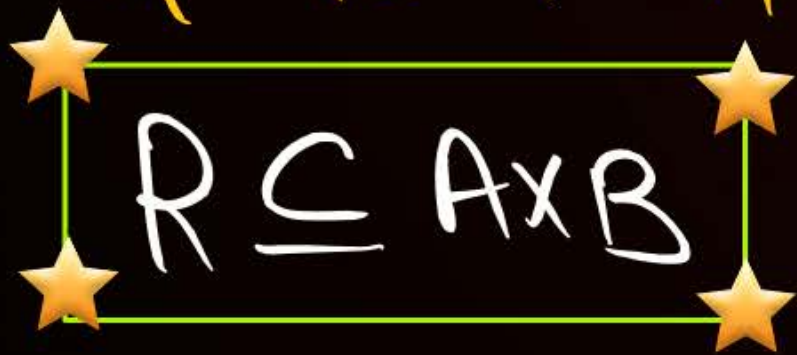
$$A = \{ \text{kohli, Vicky Kaushal, Saiy} \}$$

$$B = \{ \text{Anushka, Katrina} \}$$

$$R = \{ (a, b) : \text{a is the husband of b} \mid a \in A, b \in B \}$$

$$A \times B = \{ (\text{koh}, \text{Anu}), (\text{koh}, \text{Kat}), (\text{Vic}, \text{Anu}), (\text{Vic}, \text{Kat}), (\text{Saiy}, \text{Anu}), (\text{Saiy}, \text{Kat}) \}$$

$$R = \{ (\text{koh}, \text{Anu}), (\text{Vic}, \text{Kat}) \}$$



Ex:

$$A = \{1, 2, 3, 4\}$$

$$B = \{4, 9, 8, 16\}$$

$$n(A \times B) = 16.$$

$$R = \{(a, b) : a^2 = b \text{ and } a \in A, b \in B\}$$

$$R = \{(2, 4), (3, 9), (4, 16)\}$$

$$R: A \rightarrow B$$



Relation



A connection between 2 sets is called a Relation.

$$R \subseteq A \times B$$

-3
-1
1
2
3
4

1
2
3
4
5
6
7
8
9

Technical Definition:

Relation is a subset of cartesian product A and B.

QUESTION



A relation is defined on set of natural numbers such that

$R = \{(x, x^3) : x \text{ is a prime number less than } 10\}$. Write this relation in Roster form.

$$R: \mathbb{N} \rightarrow \mathbb{N}.$$

$$R = \{(2, 8), (3, 27), (5, 125), (7, 343)\}$$

QUESTION



If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, which of the following are relations from A to B ? Give reasons in support of your answer.

- (i) $R_1 = \{(1, 4), (1, 5), (1, 6)\}$ ✓
- (ii) $R_2 = \{(1, 5), (2, 4), (3, 6)\}$ ✓
- (iii) $R_3 = \{(1, 4), (1, 5), (3, 6), (2, 6), (3, 4)\}$ ✓
- (iv) $R_4 = \{(4, 2), (2, 6), (5, 1), (2, 4)\}$ ✗

$R_4 \not\subseteq A \times B$

$$n(A \times B) = 9$$



Total Number of Relations

$$A = \{1, 2\}$$

$$\text{no. of subsets} = 2^2 = 4$$

$$\phi, \{1\}, \{2\}, \{1, 2\}$$

$$\text{no. of proper subsets} = 2^2 - 1 = 3$$

$$n(A) = 5, n(B) = 4$$

$$n(A \times B) = 20$$

$$\text{no. of subsets} = 2^{20}$$

no. of Relations

QUESTION



If R is a relation on a finite set having n elements, then the number of relations on A is

- A** 2^n
- B** 2^{n^2}
- C** n^2
- D** n^n

$$n(A) = n, \quad n(B) = n.$$
$$n(A \times B) = n \times n = n^2$$

$$\text{no. of subsets} = \text{no. of Relations} \\ = 2^{n^2}$$



Representation of a Relation

Relation $\rightarrow$ set

- ① Roster
- ② Set-Builder Form
- ③ Arrow diagram



Domain and Range of a Relation



QUESTION



If R is a relation from set $A = \{2, 4, 5\}$ to set $B = \{1, 2, 3, 4, 6, 8\}$ defined by $xRy \Leftrightarrow x$ divides y .

- (i) Write R as a set of ordered pairs.
- (ii) Find the domain and the range of R .

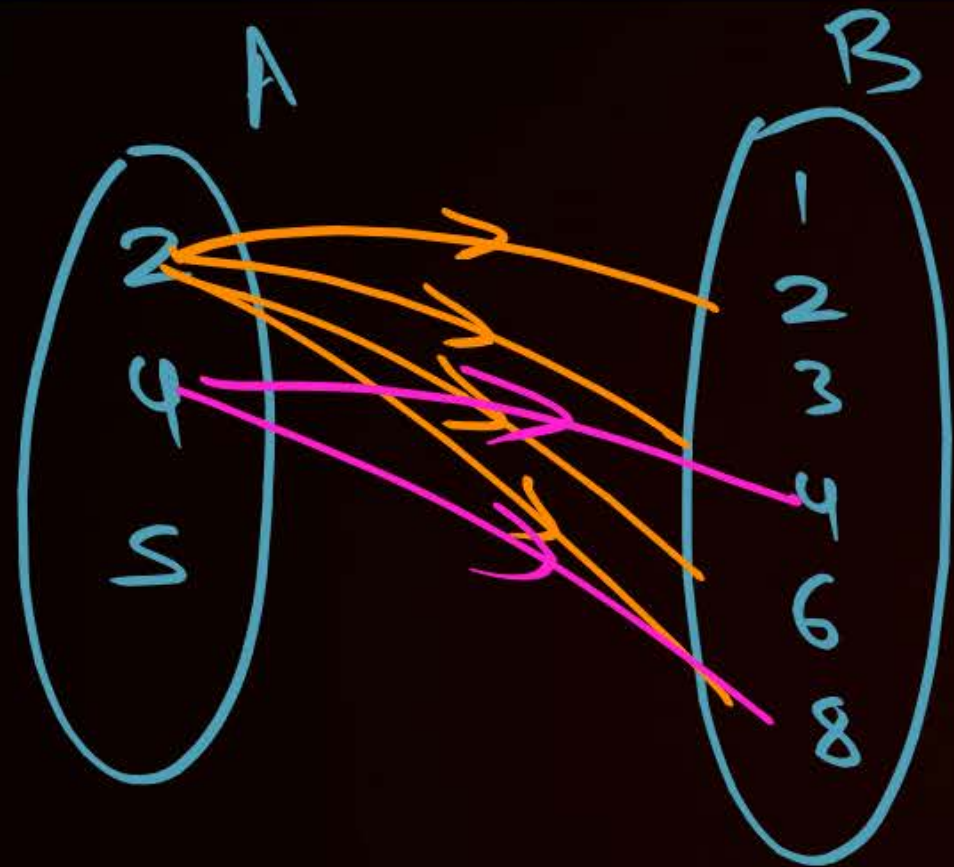
$$(i) R = \{(2, 2), (2, 4), (2, 6), (2, 8), (4, 4), (4, 8)\}$$

$$(ii) \text{ Domain} = \{2, 4\}$$

$$\text{Range} = \{2, 4, 6, 8\}$$

$$\text{Co-domain} = \text{Set } B = \{1, 2, 3, 4, 6, 8\}$$

Arrow diagram



- ★ Domain: All those elements of First set which have a image in second set.
 $D = \{2, 4\}$
- ★ Range: All those elements of second set which have a pre-image in First set.
 $R = \{2, 4, 6, 8\}$
- ★ Co-domain: whole second set
 $\{1, 2, 3, 4, 6, 8\}$

Relation

$$R = \{(a, b) : a + 3b = 12, a \in \mathbb{N}, b \in \mathbb{N}\}.$$

(i) R

(ii) Domain of R

(iii) Range of R

$$(1) R = \{(3,3), (6,2), (9,1)\}$$

(ii) $D = \{3, 6, 9\}$

(iii) $R = \{3, 2, 1\}$

Co-domain = N

QUESTION



Let A be the set of first ten natural number and let R be a relation on A defined by

$R = \{(x, y) : x \in A, y \in A \text{ and } x + 2y = 10\}$. Express R and R^{-1}

as sets of ordered pairs. Also, determine

(i) Domains of R and R^{-1}

(ii) Range of R and R^{-1} .

$$A = \{1, 2, 3, \dots, 10\}$$

$$R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$$

$$R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$$

(i) $\text{Dom of } R = \{2, 4, 6, 8\}$

$$\text{Dom of } R^{-1} = \{4, 3, 2, 1\}$$

(ii) $\text{Range of } R = \{4, 3, 2, 1\}$

$$\text{" " } R^{-1} = \{2, 4, 6, 8\}$$

Domain of Relation = Range of inverse Relation.

Range of Relation = Domain of inverse Relation.



Inverse of a Relation



Inverse Relation:

Let A, B be two sets and let R be a relation from a set A to a set B . Then, the inverse of R , denoted by R^{-1} , is a relation from B to A and is defined by $R^{-1} = \{(b, a) : (a, b) \in R\}$.



QUESTION



Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation R on set A by $R = \{(x, y) : y = x + 1\}$.

- (i) Depict this relation using an arrow diagram.
- (ii) Write down the domain, co-domain and range of R .

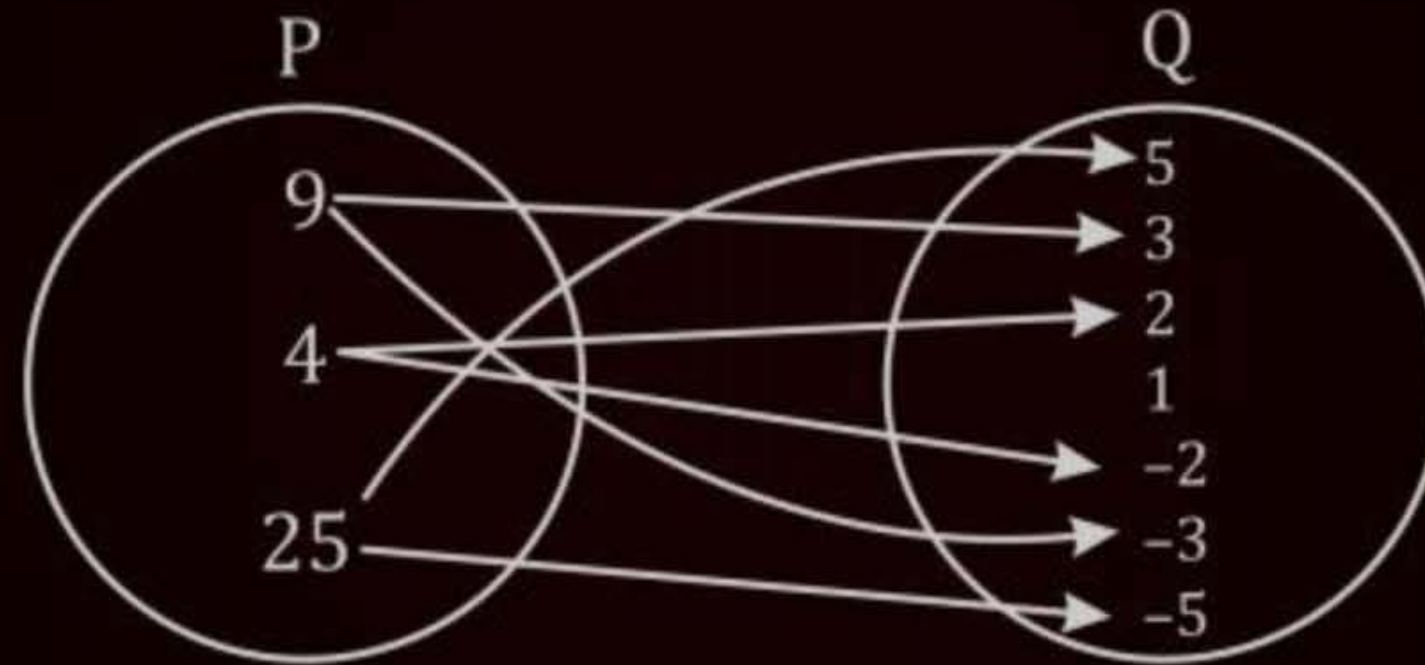
H.w

QUESTION



Fig. shows a relation R between the sets P and Q.

What is its domain and Range?



$\{9, 4, 25\}$

$\{5, 3, 2, -2, -3, -5\}$

QUESTION

$$R: \mathbb{Q} \rightarrow \mathbb{Q}$$



Rational no.

Let R be a relation on $\mathbb{Q}$ defined by $R = \{(a, b) : a, b \in \mathbb{Q} \text{ and } a - b \in \mathbb{Z}\}$.

Show that:

- (i) $(a, a) \in R$ for all $a \in \mathbb{Q}$
- (ii) $(a, b) \in R \Rightarrow (b, a) \in R$
- (iii) $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$

$$(i) (a, a) \in R, \forall a \in \mathbb{Q}$$

$$\because a - a = 0$$

$$0 \in \mathbb{Z}$$

$$(ii) (a, b) \in R \\ \because a - b \in \mathbb{Z}$$

$$\Rightarrow -(a - b) \in \mathbb{Z} \\ -a + b \in \mathbb{Z} \\ b - a \in \mathbb{Z}$$

$$(iii) (a, b) \in R \\ \Rightarrow a - b \in \mathbb{Z} \quad \text{--- (1)}$$

$$\text{Also, } (b, c) \in R$$

$$\Rightarrow b - c \in \mathbb{Z} \quad \text{--- (2)}$$

$$(1) + (2)$$

$$(a - b) + (b - c) \in \mathbb{Z}$$

$$\Rightarrow (b, a) \in R \quad a - \cancel{b} + \cancel{b} - c \in \mathbb{Z}$$

$$a - c \in \mathbb{Z}$$

$$\Rightarrow (a, c) \in R$$

QUESTION



Let R be a relation on N defined by $R = \{(a, b) : a, b \in N \text{ and } a = b^2\}$.

~~Check~~ Check:

- (i) $(a, a) \in R$ for all $a \in N$
- (ii) $(a, b) \in R \Rightarrow (b, a) \in R$
- (iii) $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$

Justify your answer in each case.

(i) $(a, a) \in R$ only for $a=1$.
 $\therefore (a, a) \notin R$ for all $a \in N$.

(ii) $(4, 2) \in R$ but $(2, 4) \notin R$
 $\therefore (a, b) \in R$ does not imply that $(b, a) \in R$.

(iii) $(16, 4) \in R$
Also, $(4, 2) \in R$
but $(16, 2) \notin R$.

$\therefore (a, b) \in R$ and $(b, c) \in R$
does not imply that $(a, c) \in R$.

QUESTION



The relation on the set $A = \{x : |x| < 3, x \in \mathbb{Z}\}$ is defined by $R = \{(x, y) : y = |x|, x \neq -1\}$. Then the number of elements in the power set of R is

A 32

B 16

C 8

D 64

QUESTION



Let $n(A) = m$ and $n(B) = n$. Then the total number of non-empty relations that can be defined from A to B is

- A** m^n
- B** $n^m - 1$
- C** $mn - 1$
- D** $2^{mn} - 1$

$$R: A \rightarrow B$$

$$n(A) = m$$

$$n(B) = n$$

$$n(A \times B) = mn$$

$$\text{no. of relations} = 2^{mn}$$

QUESTION



$A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation R from A to B by $R = \{(x, y) : \text{the difference between } x \text{ and } y \text{ is odd, } x \in A, y \in B\}$. Write R in Roster form.

$$R = \{(1, 4), (1, 6), (2, 9), (3, 4), (3, 6), (5, 4), (5, 6)\}$$

QUESTION



A relation R is defined on the set Z of integers as $(x, y) \in R \Leftrightarrow x^2 + y^2 = 25$.



ASSERTION-REASON BASED MCQS

Each of the following examples contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason) and has the following four choices (a), (b), (c) and (d), only one of which is the correct answer. Mark the correct choice.

- (a) Statement-1 and Statement-2 are true; Statement-2 is a correct explanation for Statement-1.
- (b) Statement-1 and Statement-2 are true; Statement-2 is not a correct explanation for Statement-1.
- (c) Statement-1 is True, Statement-2 is False.
- (d) Statement-1 is False, Statement - 2 is True.

QUESTION



Statement-1 (A): Let A and B be two sets such that $n(A \cap B') = 4$, $n(A' \cap B') = 12$ and $n(A \cap B) = 8$, then $n(A \times B) = 240$.

Statement-2 (R): For any two non-empty sets A and B , $n(A)$ and $n(B)$ are divisors of $n(A \times B)$.



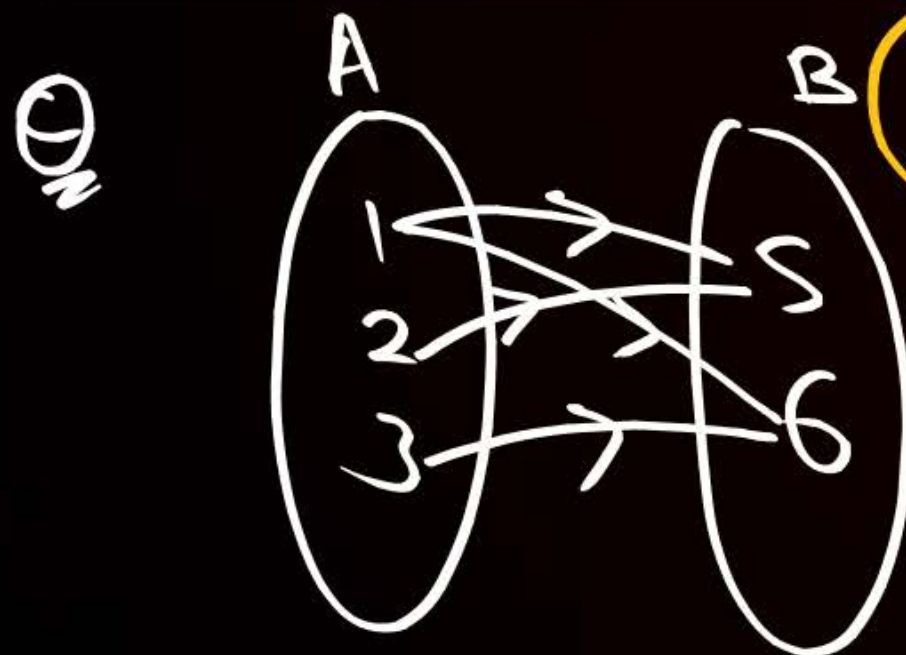
What is a function

Special relation.

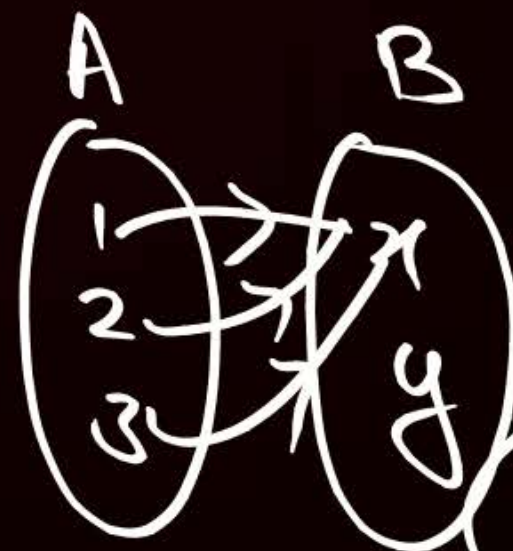


A relation R from a set A to set B is called a function.

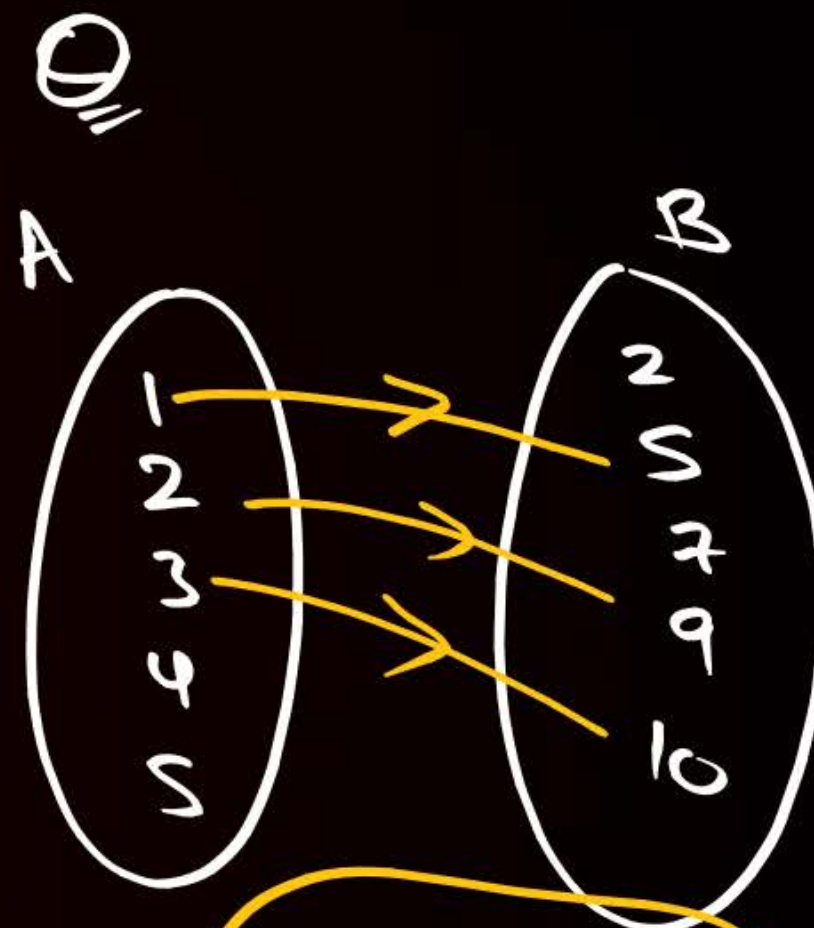
- (i) Every element of set A has an image in set B .
- (ii) No element of set A has more than one image in set B .



not a function



Function!



not a function

NOTE :

- (i) Converging rays are allowed but diverging rays are not allowed.
- (ii) Complete set A must be domain.

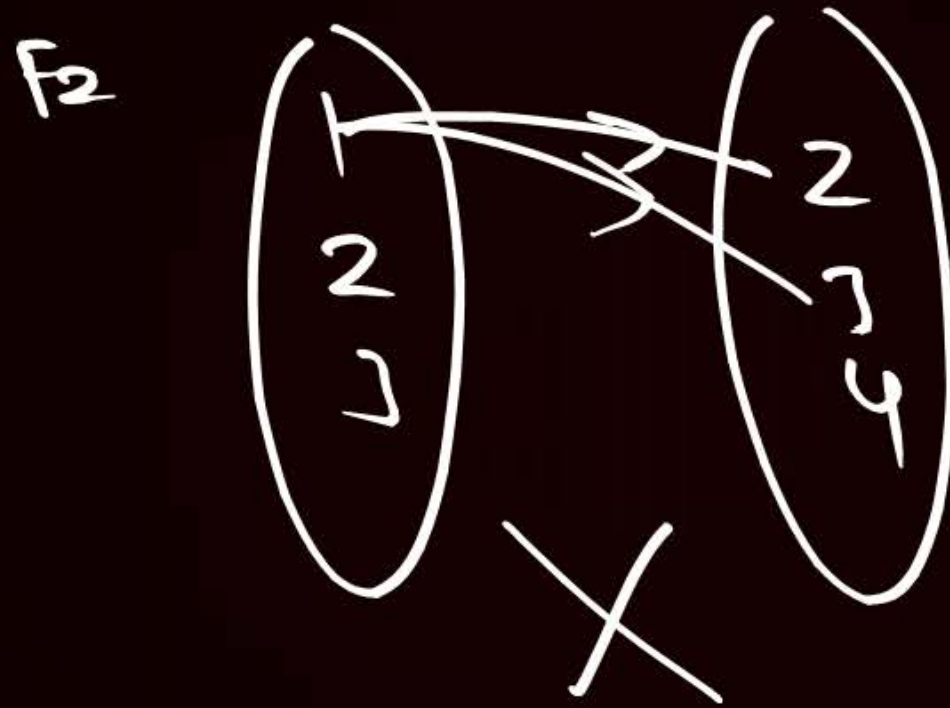
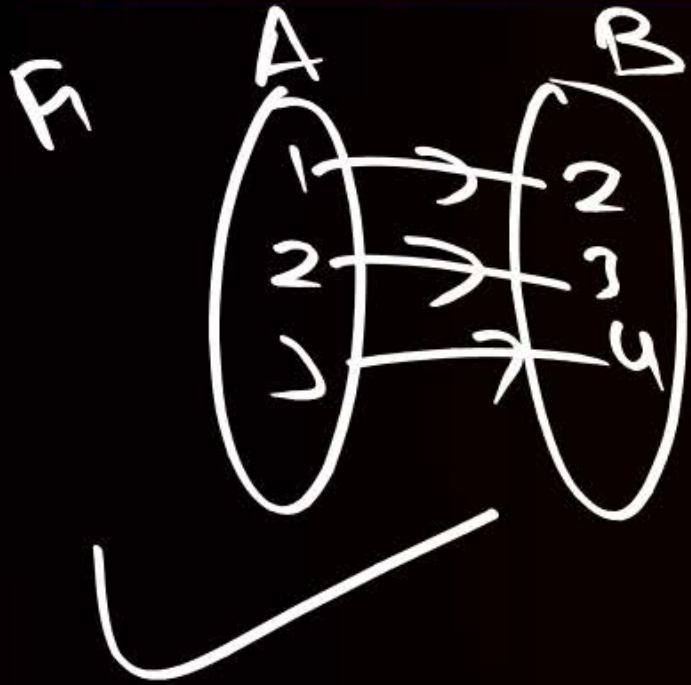
QUESTION



Relations

Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and f_1, f_2 and f_3 be three subset of $A \times B$ as given below:

$f_1 = \{(1, 2), (2, 3), (3, 4)\}$, $f_2 = \{(1, 2), (1, 3), (2, 3), (3, 4)\}$, $f_3 = \{(1, 3), (2, 4)\}$.




 If $(a, b) \in f$, then b is called the image of a under f .

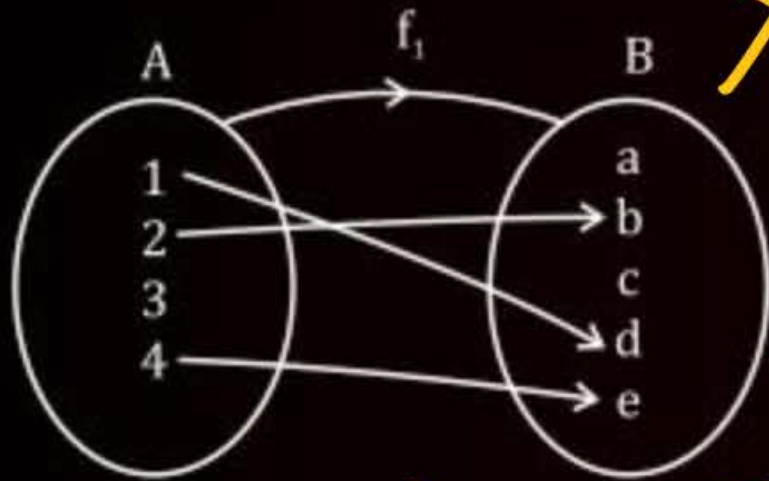
$$(2, 5) \in F$$

QUESTION

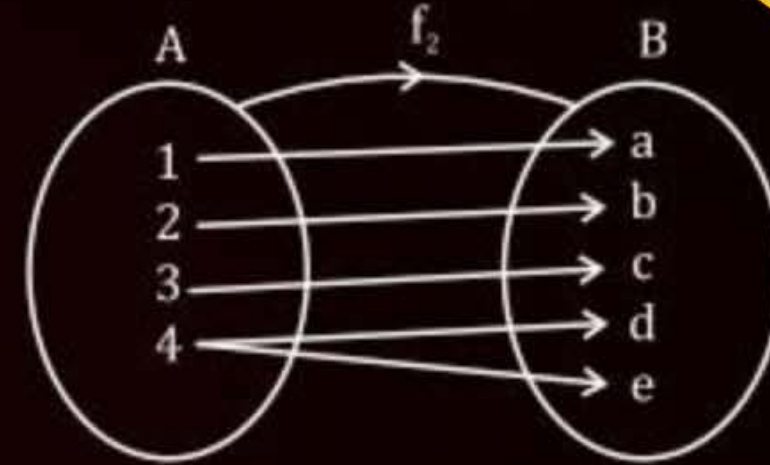


Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d, e\}$ be two set and let f_1, f_2, f_3 and f_4 be rules associating elements of set A to elements of set B as shown in fig. 1 to fig. 4.

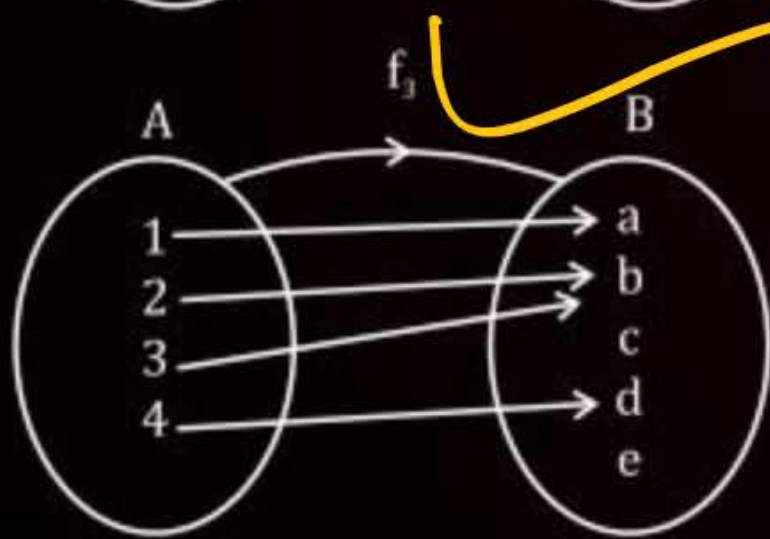
1.



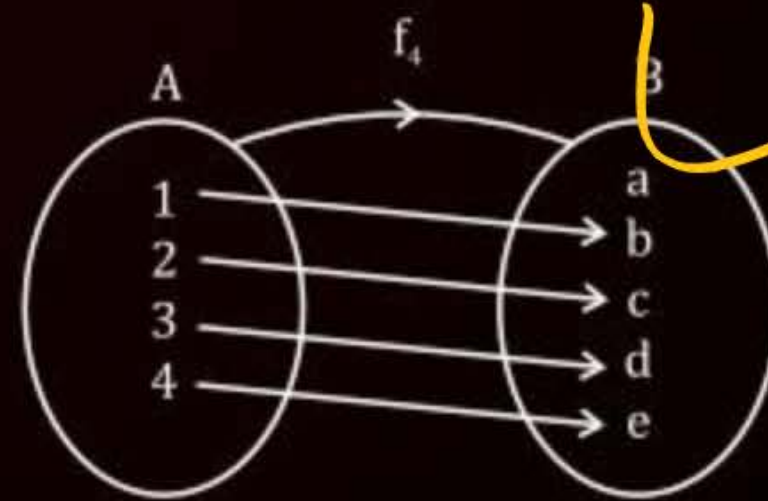
2.



3.



4.



QUESTION



$$x \in \{1, 2, 3, 4\}$$
$$y \in \{1, 2, 3, 4\}$$

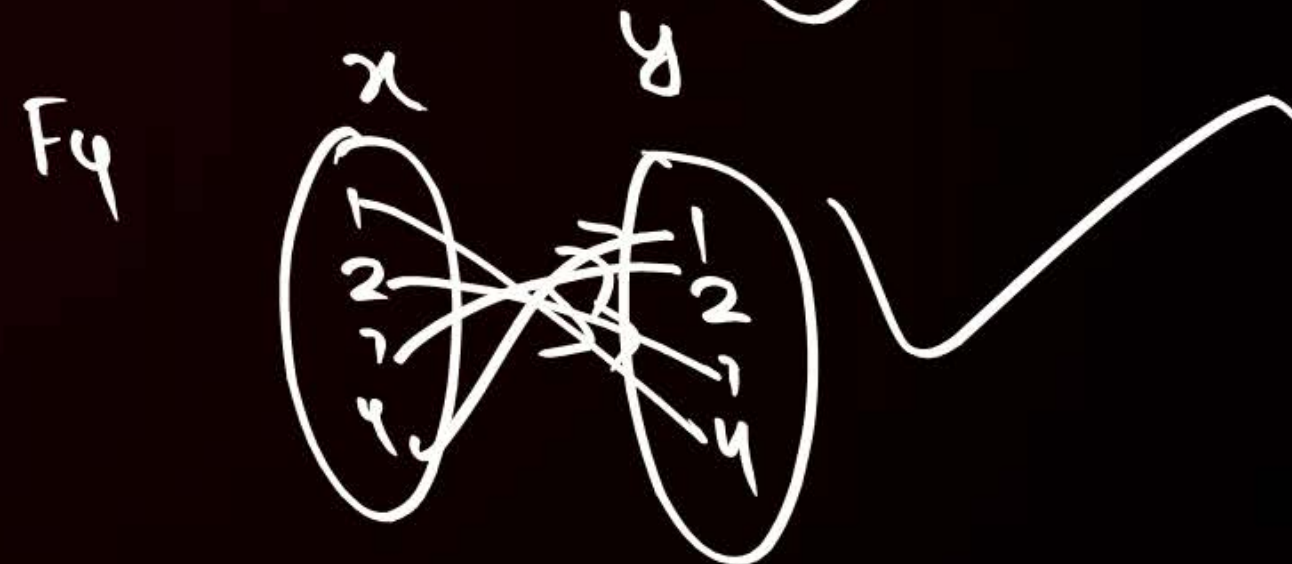
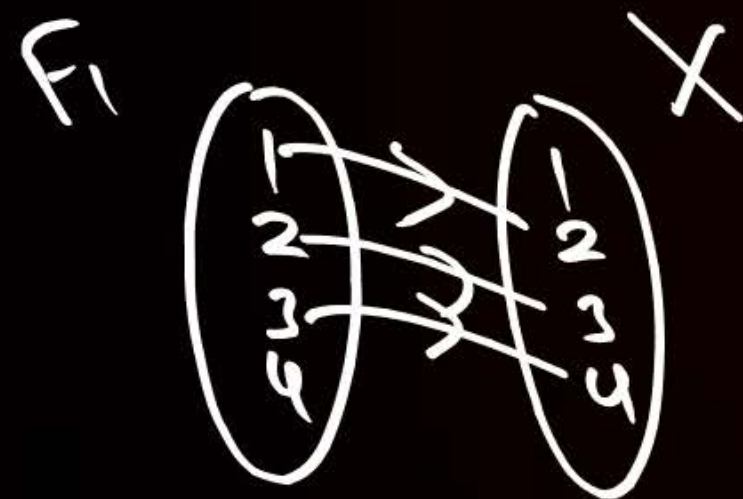
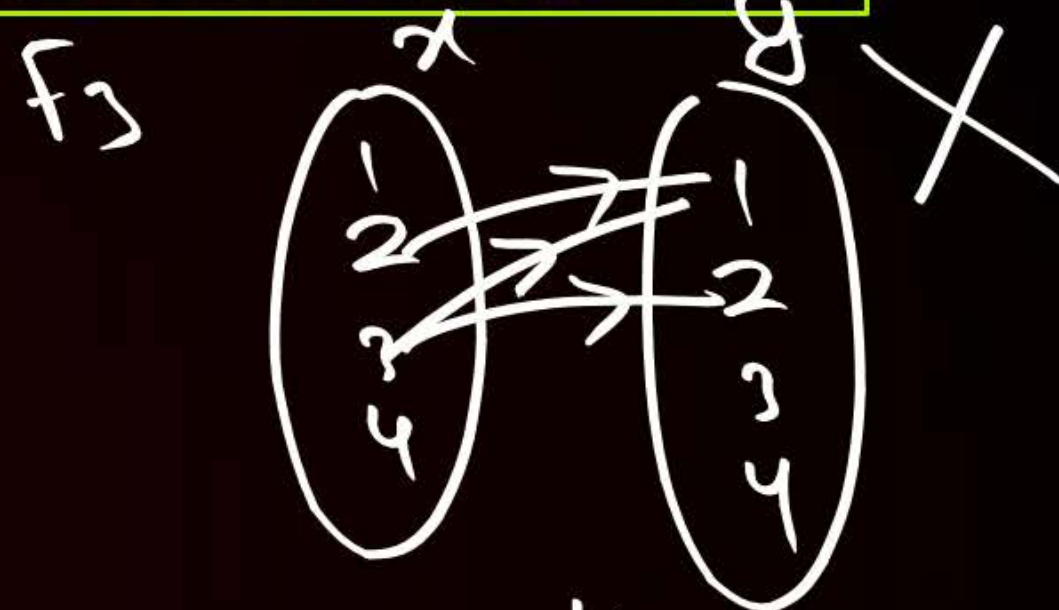
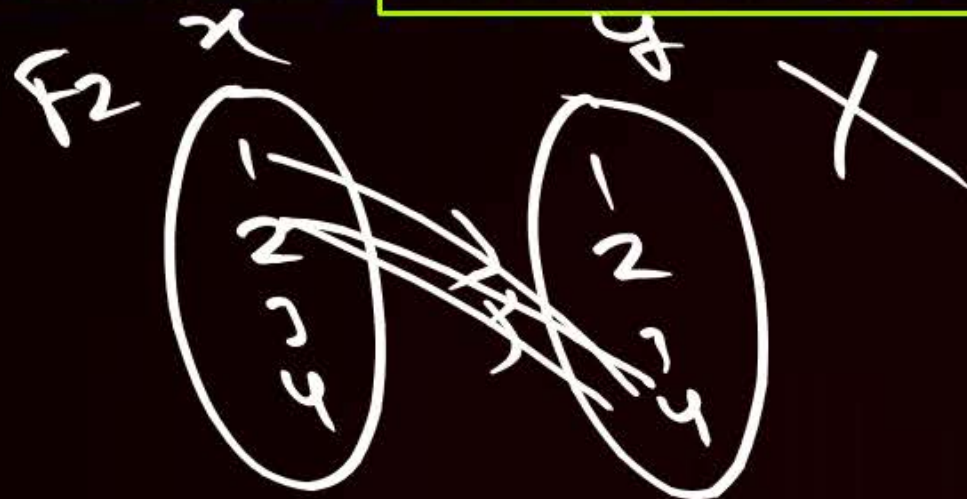
If $x, y \in \{1, 2, 3, 4\}$, then which of the following are functions in the given set?

$$f_1 \{(x, y) : y = x + 1\}$$

$$f_2 \{(x, y) : x + y > 4\}$$

$$f_3 \{(x, y) : y < x\}$$

$$f_4 \{(x, y) : x + y = 5\}$$



QUESTION



Let $A = \{-2, -1, 0, 1, 2\}$ and $f: A \rightarrow \mathbb{Z}$ be given by $f(x) = x^2 - 2x - 3$. find

- The range of
- Pre-images of 6, -3 and 5.

$$f(-2) = (-2)^2 - 2(-2) - 3$$

$$f(-2) = 4 + 4 - 3$$

$$f(-2) = 5$$

$$f(-1) = (-1)^2 - 2(-1) - 3$$

$$= 1 + 2 - 3$$

$$f(-1) = 0$$

$$f(0) = (0)^2 - 2(0) - 3$$

$$f(0) = -3$$

$$f(1) = (1)^2 - 2(1) - 3$$

$$= 1 - 2 - 3$$

$$= -4$$

$$f(1) = -4$$

$$f(2) = (2)^2 - 2(2) - 3$$

$$= 4 - 4 - 3$$

$$f(2) = -3$$

$$F = \{ (x, f(x)) : x \in A \text{ and } f(x) \in \mathbb{Z} \}$$

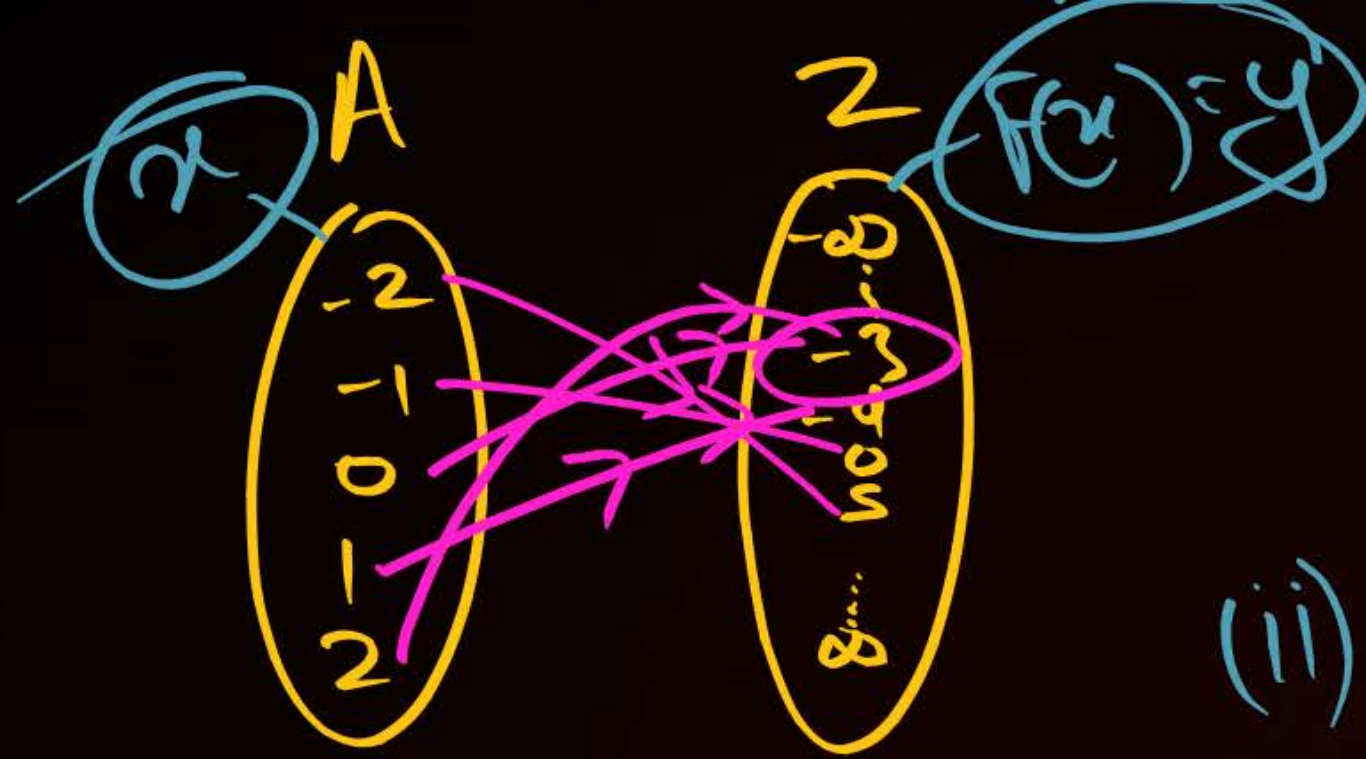
$$F = \{ (-2, 5), (-1, 0), (0, -3), (1, -4), (2, -3) \}$$

$$\text{Range} = \{ 5, 0, -3, -4 \}$$

Function.

y

$x \in A$
 $f(x), y \in \mathbb{Z}$



Pre-image of $-3 = \{0, 2\}$ ~~X~~

Pre-image of $5 = \{-2\}$

(ii) $f(x) = 6, 7, 5$

$f(x) = 6$

$x^2 - 2x - 3 = 6$

$x^2 - 2x - 9 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-9)}}{2(1)}$

$x = \frac{2 \pm \sqrt{40}}{2}$

6 has no pre-image

(i) $f(x) = -3$

$x^2 - 2x - 3 = -3$

$x^2 - 2x + 0 = 0$

$x^2 - 2x = 0$

$x(x-2) = 0$

$x = 0, x = 2$

$f(x) = 5$

$x^2 - 2x - 3 = 5$

$x^2 - 2x - 8 = 0$

$x^2 - 4x + 2x - 8 = 0$

$x(x-4) + 2(x-4) = 0$

$(x-4)(x+2) = 0$

~~$x = 4, -2$~~

QUESTION



$g(x) = x^2 - 10x - 18$. Determine the image of -2 and the pre-image of 6 .

$$g(x) = y$$



$$\begin{aligned} g(-2) &= (-2)^2 - 10(-2) - 18 \\ &= 4 + 20 - 18 \\ &= 4 + 2 \end{aligned}$$

$$g(-2) = 6$$

$$\text{image of } -2 = 6$$

$$\begin{aligned} g(x) &= 6 \\ x^2 - 10x - 18 &= 6 \\ x^2 - 10x - 24 &= 0 \end{aligned}$$

$$x^2 - 12x + 2x - 24 = 0$$

$$x(x-12) + 2(x-12) = 0$$

$$(x-12)(x+2) = 0$$

$$x = 12, x = -2$$

Pre-image of 6 are -2 and 12



Description of a function

Special relation $\rightarrow$ Set $\rightarrow$ Ordered Pair $\rightarrow$ Cartesian Product



Let $f : A \rightarrow B$ be a function such that the set A consists of a finite number of elements. Then, $f(x)$ can be described by listing the values which it attains at different points of its domain.

Example, if $A = \{-1, 1, 2, 3\}$ and B is the set of real numbers, then a function $f : A \rightarrow B$ can be described as $f(-1) = 3$, $f(1) = 0$, $f(2) = 3/2$ and $f(3) = 0$. In case, A is an infinite set, then f cannot be described by listing the images at points in its domain. In such cases functions are generally described by a formula.

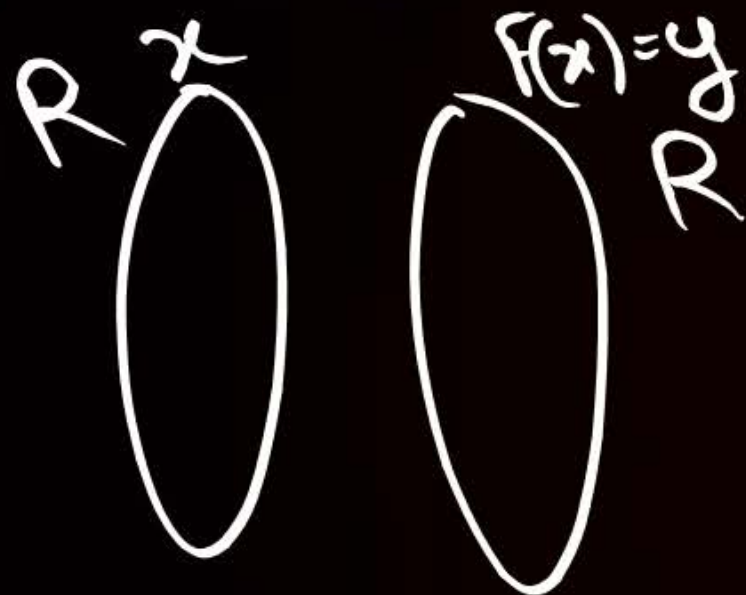
For example, $f : \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2 + 1$ or $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = e^x$ etc.

$$f : \{(-1, 3), (1, 0), (2, \frac{3}{2}), (3, 0)\}$$

QUESTION



If a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x - 2, & x < 0 \\ 1, & x = 0 \\ 4x + 1, & x > 0 \end{cases}$. Find $f(1)$, $f(-1)$, $f(0)$, $f(2)$.



$$f(1) = 4(1) + 1$$

$$f(1) = 5$$

$$f(-1) = 3(-1) - 2$$

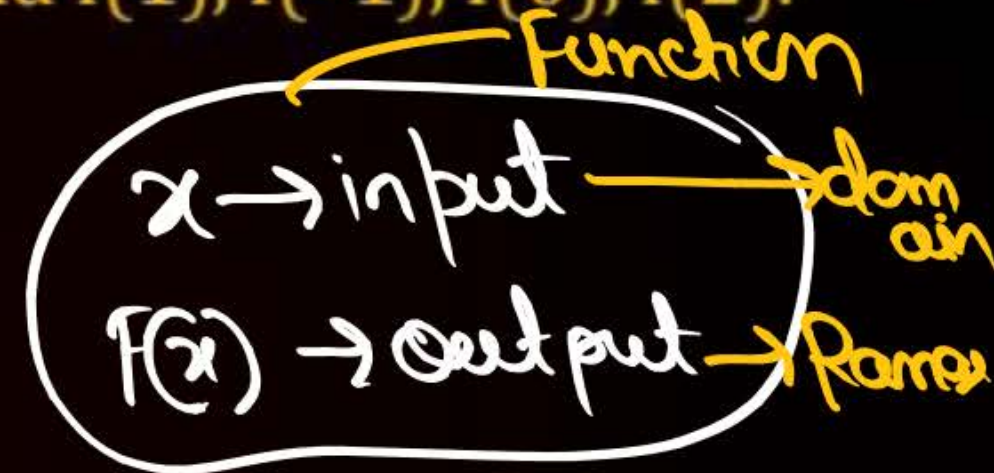
$$= -3 - 2$$

$$f(-1) = -5$$

$$f(0) = 1$$

$$f(2) = 4(2) + 1$$

$$f(2) = 9$$



QUESTION

If $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ -1, & \text{if } x \notin \mathbb{Q} \end{cases}$

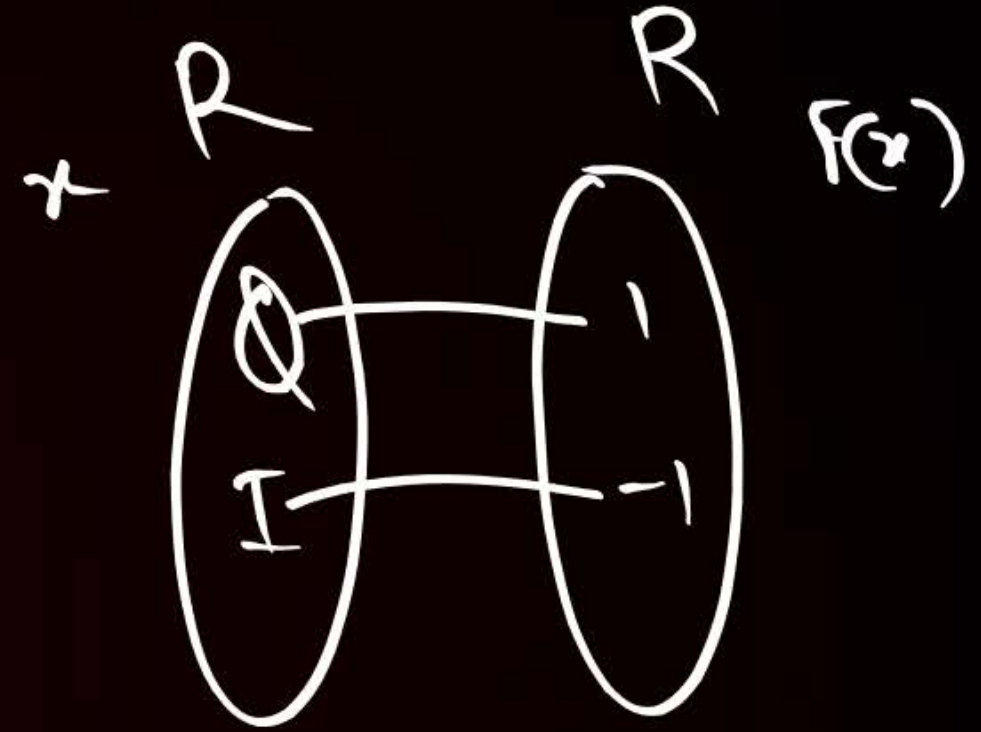
Find

- (i) $f(1/2), f(\pi), f(\sqrt{2})$
- (ii) Range of f
- (iii) Pre-image of 1 and -1.

(i) $f(1/2) = 1, f(\pi) = -1, f(\sqrt{2}) = -1$

(ii) Range = Values of $f(x) = \{1, -1\}$

(iii) Pre image of 1 = $\mathbb{Q}$
Pre image of -1 = Irrational nos.



QUESTION



Relation.

The ~~function~~ f is defined by $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 3x, & 3 \leq x \leq 10 \end{cases}$.

The relation g is defined by $g(x) = \begin{cases} x^2, & 0 \leq x \leq 2 \\ 3x, & 2 \leq x \leq 10 \end{cases}$.

Show that f is a function and g is not a function.

$$F = \{(0,0), (1,1), (2,4), (3,9), (4,12), (5,15), (6,18), (7,21), (8,24), (9,27), (10,30)\}$$

$$g = \{(0,0), (1,1), (2,4), (2,6), (3,9), (4,12), (5,15), (6,18), (7,21), (8,24), (9,27), (10,30)\}$$

2 has more than one image ' g ', so g is not a function.

QUESTION



If $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 1$, then find $f^{-1}\{17\}$ and $f^{-1}\{-3\}$.

$$f(x) = x^2 + 1$$

$$17 = x^2 + 1$$

$$16 = x^2$$

$$\pm \sqrt{16} = x$$

$$\pm 4 = x$$

pre-image of 17

$$f^{-1}\{17\} = \{4, -4\}$$

How



Equal Function

Function ke liye
Pehla set pura
domain hota



Definition :

Two functions f and g are said to be equal iff

- (i) Domain of f = domain of g
- (ii) Co-domain of f = Co-domain of g and
- (iii) $f(x) = g(x)$ for every x belonging to their common domain.

If two function f and g are equal, then we write $f = g$.

QUESTION



Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$ be function defined by $f = \{(n, n^2) : n \in \mathbb{Z}\}$ and, $g = \{(n, |n^2|) : n \in \mathbb{Z}\}$. Show that $f = g$.

- ① domain same.
- ② co-domain same.
- ③ $f(n) = n^2$ and $g(n) = |n^2|$ both are equal.

so $f = g$

$$|n^2| = n^2$$

$$n = -5$$

L.H.S

$$|(-5)^2|$$

$$|25| = 25$$

R.H.S

$$(-5)^2$$

$$= 25$$

$$\text{Mod } 3 = |3| = 3$$

$$|4| = 4$$

$$|-3| = 3$$

$$|-5| = 5$$

$$|0| = 0$$

QUESTION



Let f be defined by $f(x) = x - 4$ and g be defined by $g(x) = \begin{cases} \frac{x^2 - 16}{x + 4}, & x \neq -4 \\ \lambda, & x = -4 \end{cases}$

Find λ such that $f(x) = g(x)$ for all x .

$$f(x) = x - 4$$

$$f(-4) = -4 - 4$$

$$f(-4) = -8$$

$$\therefore f(x) = g(x)$$

$$f(-4) = g(-4)$$

$$g(x) \begin{cases} \frac{x^2 - 16}{x + 4}, & x \neq -4 \\ \lambda, & x = -4 \end{cases}$$

$$-8 = \lambda$$



Real Valued Function

Real no.

$$B \subseteq \mathbb{R}$$

A function $f : A \rightarrow B$ is called a real valued function, if B is a subset of $\mathbb{R}$ (set of all real numbers).

If A and B both are subsets of $\mathbb{R}$, then f is called a real function.

$$\text{If } A, B \subseteq \mathbb{R}$$

$F \rightarrow \text{Real Function}$



QUESTION



If $f(x) = \frac{x-1}{x+1}$, $x \neq -1$, then show that $f(f(x)) = -\frac{1}{x}$, provided that $x \neq 0$.

$$f(x) = \frac{x-1}{x+1}$$

$$f(2) = \frac{2-1}{2+1}$$

$$f(3) = \frac{3-1}{3+1}$$

$$f(y) = \frac{y-1}{y+1}$$

$$f(a) = \frac{a-1}{a+1}$$

$$f(f(x)) = \frac{f(x)-1}{f(x)+1}$$

$$\begin{aligned}
 &= \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} \\
 &= \frac{(x-1) - (x+1)}{(x-1) + (x+1)} \\
 &= \frac{x-1-x-1}{x-x+x+1} = \frac{-2}{2x} = -\frac{1}{x}
 \end{aligned}$$



Domain

→ All possible values of 'x' for which

$f(x)$ is defined.

Real no.

QUESTION



$$f(x) = x^3$$

$$f(x) = x^3$$

$$x \in (-\infty, \infty)$$

or

$$x \in \mathbb{R}$$

QUESTION

$$f(x) = \frac{1}{x-3}$$

$$x-3 \neq 0$$

$$x \neq 3$$

$$D: x \in \mathbb{R} - \{3\}$$

QUESTION

$$f(x) = \sqrt{x}$$

$$x \in [0, \infty)$$



QUESTION

$$f(x) = \sqrt{x-3}$$

A $[0, \infty)$

B $[3, \infty)$

C $(3, \infty)$

D None

$$\sqrt{x-3}$$

$$x-3 \geq 0$$

$$x \geq 3$$

$$\text{Ans: } [3, \infty)$$

$\sqrt{\text{negativ no.}}$ = not a
real no.

QUESTION



$$f(x) = \frac{1}{\sqrt{x-3}}$$

$$x-3 > 0$$

$$x > 3$$

$$\text{Ans: } (3, \infty)$$

QUESTION



$$f(x) = \frac{1}{x+7}$$

$$x+7 \neq 0$$

$$x \neq -7$$

$$\text{Ans: } R - \{-7\}$$

QUESTION



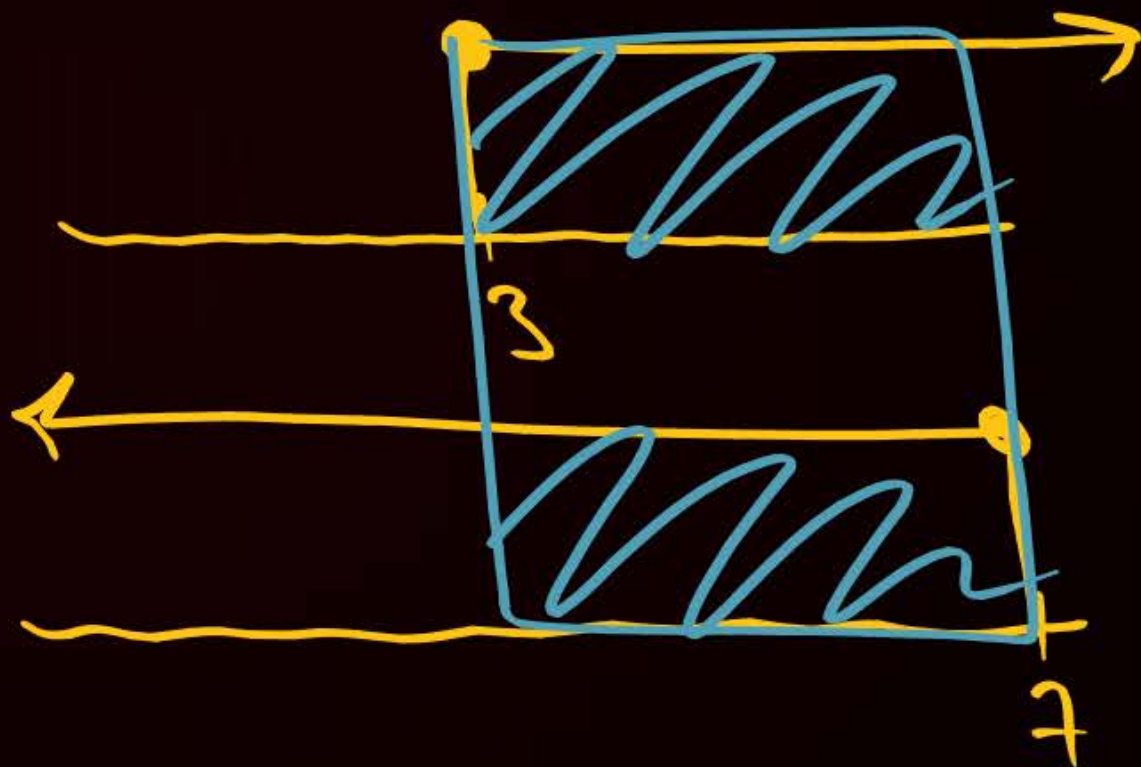
$$f(x) = \sqrt{x-3} + \sqrt{7-x}$$

$$x-3 \geq 0, \quad 7-x \geq 0$$

$$x \geq 3$$

$$7 \geq x$$

Ans: $[3, 7]$



Q domain: $f(x) = \sqrt{9-x^2}$



$$9-x^2 \geq 0$$

$$3^2-x^2 \geq 0$$

$$(3-x)(3+x) \geq 0$$

$$-(-3+x)(3+x) \geq 0$$

$$(x-3)(x+3) \leq 0$$

$$x-3=0$$

$$\Rightarrow x=3$$

$$x+3=0$$

$$\Rightarrow x=-3$$

Ams: $[-3, 3]$





Range of a Function

$$y = f(x) = x^2$$

$$f(x) = y = x^2$$

Domain: $\mathbb{R}$

$$y_{\min} = x^2_{\min} \rightarrow 0$$

$$y_{\min} = 0$$

$$y_{\max} = x^2_{\max} \rightarrow \infty$$

$$y_{\max} = \infty$$

Range: possible values of $f(x) = y$

Ans: $[0, \infty)$



QUESTION

Range



$$f(x) = 3 + 5x^2$$

$$f(x)_{\max} = x^2_{\max} = \infty$$

$$f(x)_{\min} = x^2_{\min} = 0$$

$$f(x)_{\max} = \infty$$

$$f(x)_{\min} = 3$$

Ans. $[3, \infty)$

QUESTION

Range



$$7 - x^2 = y$$

$$f(x) = y = 7 - x^2$$

$$y_{\max} = x^2_{\min} = 0$$

$$y_{\max} = 7$$

$$y_{\min} = x^2_{\max} = \infty$$

$$y_{\min} = -\infty$$

$$(-\infty, 7]$$

Ans.

QUESTION



Find the domain and range of $f(x) = \frac{3-x}{3x+4}$

x ki values

$f(x)$ ki values

$$3x+4 \neq 0$$

$$3x \neq -4$$

$$x \neq -4/3$$

$$\text{Domain: } \mathbb{R} - \left\{-\frac{4}{3}\right\}$$

$$y = \frac{3-x}{3x+4}$$

$$y(3x+4) = 3-x$$

$$3xy + 4y = 3-x$$

$$3xy + x = 3 - 4y$$

$$x(3y+1) = 3-4y$$

$$x = \frac{3-4y}{3y+1}$$

$$3y+1 \neq 0$$

$$3y \neq -1$$

$$y \neq -1/3$$

$$\text{Range: } \mathbb{R} - \left\{-\frac{1}{3}\right\}$$

QUESTION



Let $f = \left\{ \left(x, \frac{x^2}{1+x^2} \right) : x \in \mathbb{R} \right\}$ be a function from $\mathbb{R}$ into $\mathbb{R}$. Determine the range of f .

$$f(x) = \frac{x^2}{1+x^2}$$

$$y = \frac{x^2}{1+x^2}$$

$$y(1+x^2) = x^2$$

$$y + yx^2 = x^2$$

$$y = x^2 - yx^2$$

$$y = x^2(1-y)$$

$$x^2 = \frac{y}{1-y}$$

$$x = \pm \sqrt{\frac{y}{1-y}}$$

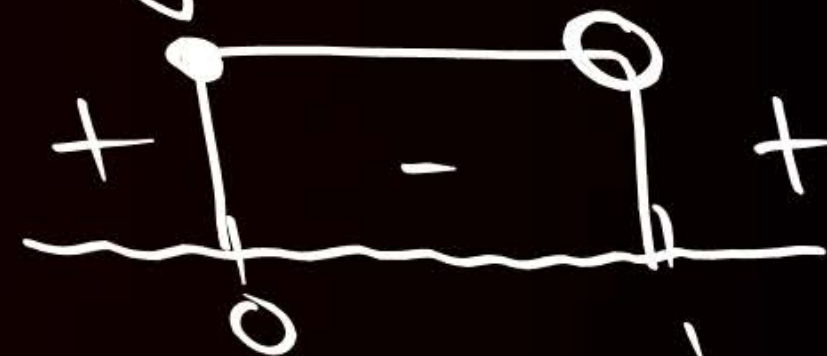
$$\frac{y}{1-y} \geq 0$$

$$\left(\frac{y}{1-y} \right) \geq 0$$

$$\frac{y}{1-y} \geq 0$$

$$\frac{y}{1-y} \leq 0$$

$$\begin{aligned} y &= 0 \\ y-1 &= 0 \end{aligned}$$



Ans: $[0, 1)$

QUESTION

$$y = \frac{x-3}{2x+1}$$

Range: $\rightarrow$ Comment



QUESTION



Domain

Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined, respectively by $f(x) = x + 1$, $g(x) = 2x - 3$. Find $f + g$, $f - g$ and $\frac{f}{g}$.

[NCERT]

$$(i) \quad f(x) + g(x) = x + 1 + 2x - 3 = 3x - 2$$

$$(ii) \quad f(x) - g(x) = (x + 1) - (2x - 3) = x + 1 - 2x + 3 = -x + 4$$

$$(iii) \quad \frac{f(x)}{g(x)} = \frac{x + 1}{2x - 3}$$

$$D: \mathbb{R} - \left\{ \frac{3}{2} \right\}$$

D: $\mathbb{R}$

QUESTION



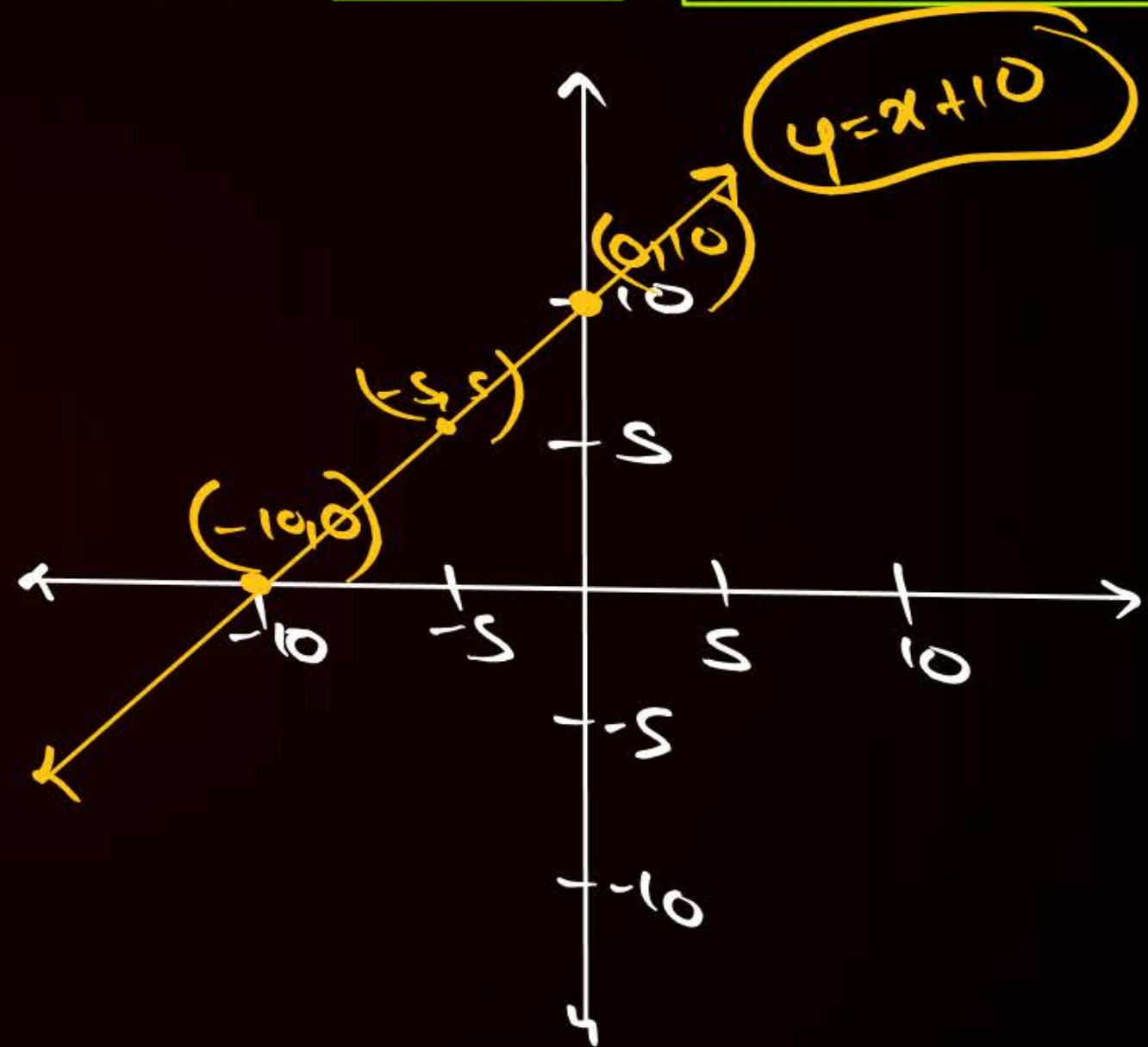
Let $\mathbb{R}$ be the set of real numbers. Define the real function $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = x + 10$ and sketch the graph of this function.

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x + 10$$

$$y = x + 10$$

x	0	-10	-5
y	10	0	5

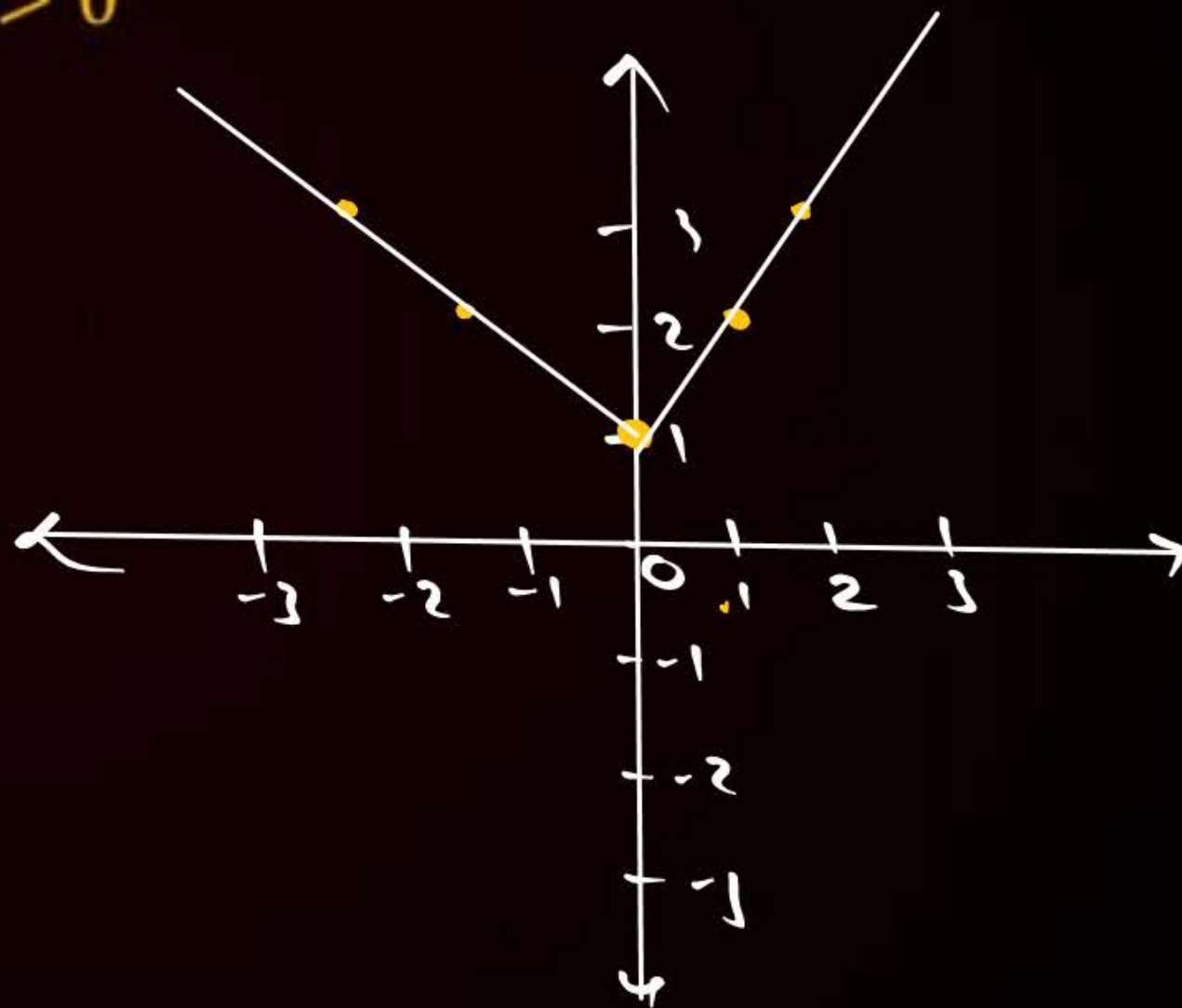


QUESTION



Ex The function f is defined by $f(x) = \begin{cases} 1-x, & x < 0 \\ 1, & x = 0 \\ x+1, & x > 0 \end{cases}$ Draw the graph of $f(x)$.

x	0	-1	1	-2	2
$f(x)$	1	2	2	3	3





SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



CONSTANT FUNCTION

If k is a fixed real number, then a function $f(x)$ given by $f(x) = k$ for all $x \in \mathbb{R}$ is called a constant function.

Ex:

$$f(x) = 2$$

$$f(x) = -3$$

$$f(x) = 100$$

$$f(x) = 3/2$$

Domain: $\mathbb{R}$

Ex:

$$f(x) = 2$$

→ A line parallel to x-axis.





SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



IDENTITY FUNCTION The function that associates each real number to itself is called the identity function and is usually denoted by I .

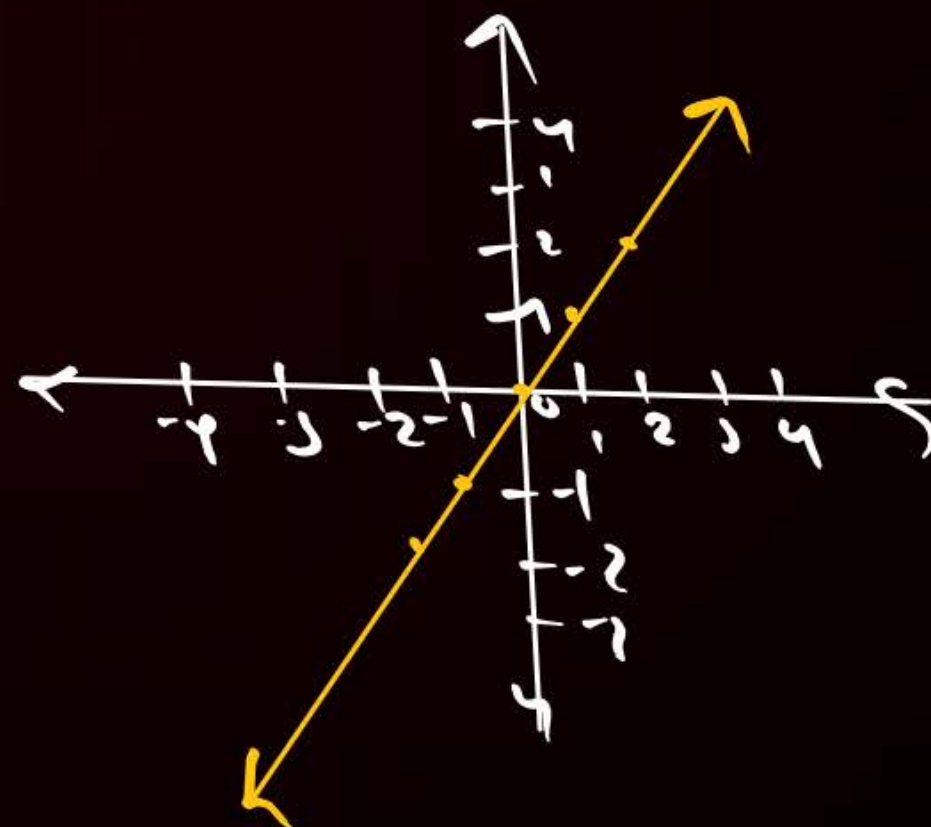
Thus, the function $I: \mathbb{R} \rightarrow \mathbb{R}$ defined by $I(x) = x$ for all $x \in \mathbb{R}$ is called the identity function.

$$f(x) = x$$

x	1	-1	0	2	-2
y	1	-1	0	2	-2

Domain: $\mathbb{R}$

Range: $\mathbb{R}$





SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



RECIPROCAL FUNCTION

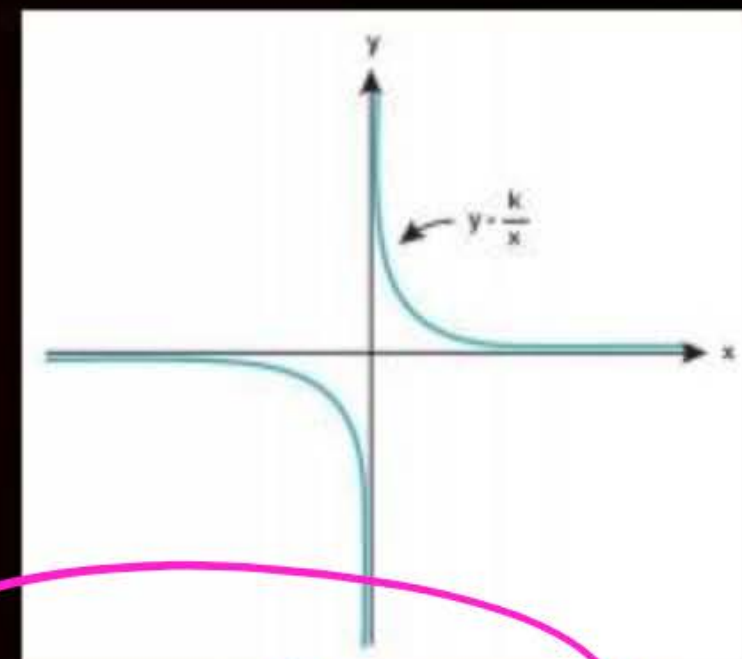
The function that associates a real number x to its reciprocal $1/x$ is called the reciprocal function. Since $1/x$ is not defined for $x = 0$. So, we define the reciprocal function as follows:

DEFINITION The function $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is called the reciprocal function.

Clearly, domain of the reciprocal function is $\mathbb{R} - \{0\}$ and its range is also $\mathbb{R} - \{0\}$.

We observe that the sign of $\frac{1}{x}$ is same as that of x and $\frac{1}{x}$ decreases with the increase in x . So, the graph of $f(x) = \frac{1}{x}$ is as shown in

$$f(x) = \frac{1}{x}$$



$$D, R: \mathbb{R} - \{0\}$$



SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



SQUARE FUNCTION The function that associates a real number x to its square i.e. x^2 is called the square function. Since x^2 is defined for all $x \in \mathbb{R}$. So, we define the square function as follows:

DEFINITION The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is called the square function.

$$f(x) = x^2$$

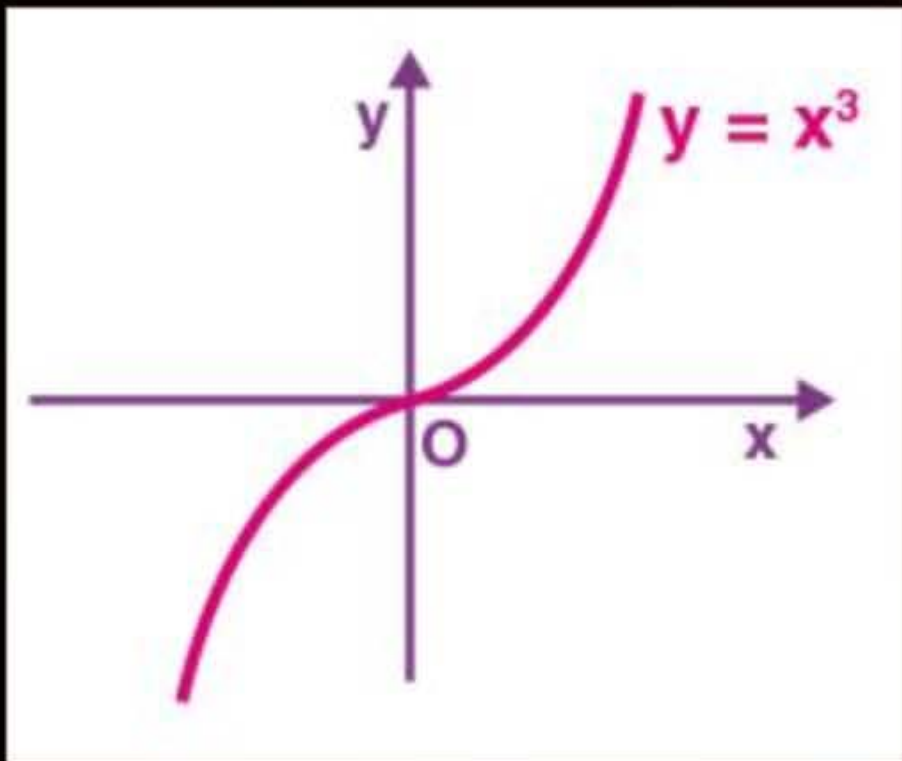


SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



CUBE FUNCTION The function that associate a real number x to its cube is called the cube function. We observe that x^3 is meaningful for all $x \in \mathbb{R}$. So, we define the cube function as follows:

DEFINITION The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$ is called the cube function.



$$f(x) = x^3$$

x	0	1	2	-2
y	0	1	8	-8



SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



MODULUS FUNCTION

The function $f(x)$ defined by $f(x) = |x| = \begin{cases} x, & \text{when } x \geq 0 \\ -x, & \text{when } x < 0 \end{cases}$ is called the modulus function.

It is also called the absolute value function.

Domain: $\mathbb{R}$
Range: $[0, \infty)$

$$y = |x|$$

$$|2| = 2 \quad |-4| = 4$$

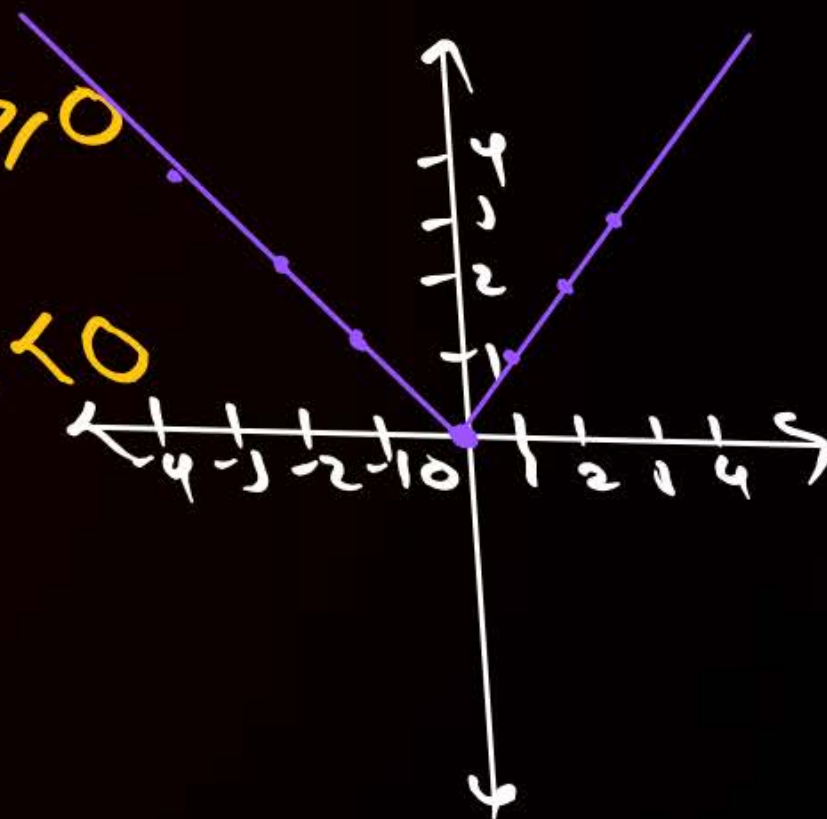
$$|-2| = 2 \quad |-0.75| = 0.75$$

$$|-3| = 3$$

$$f(x) = |x|$$

$$x, x \geq 0$$
$$-x, x < 0$$

x	0	1	-1	2	-2
y	0	1	1	2	2





SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



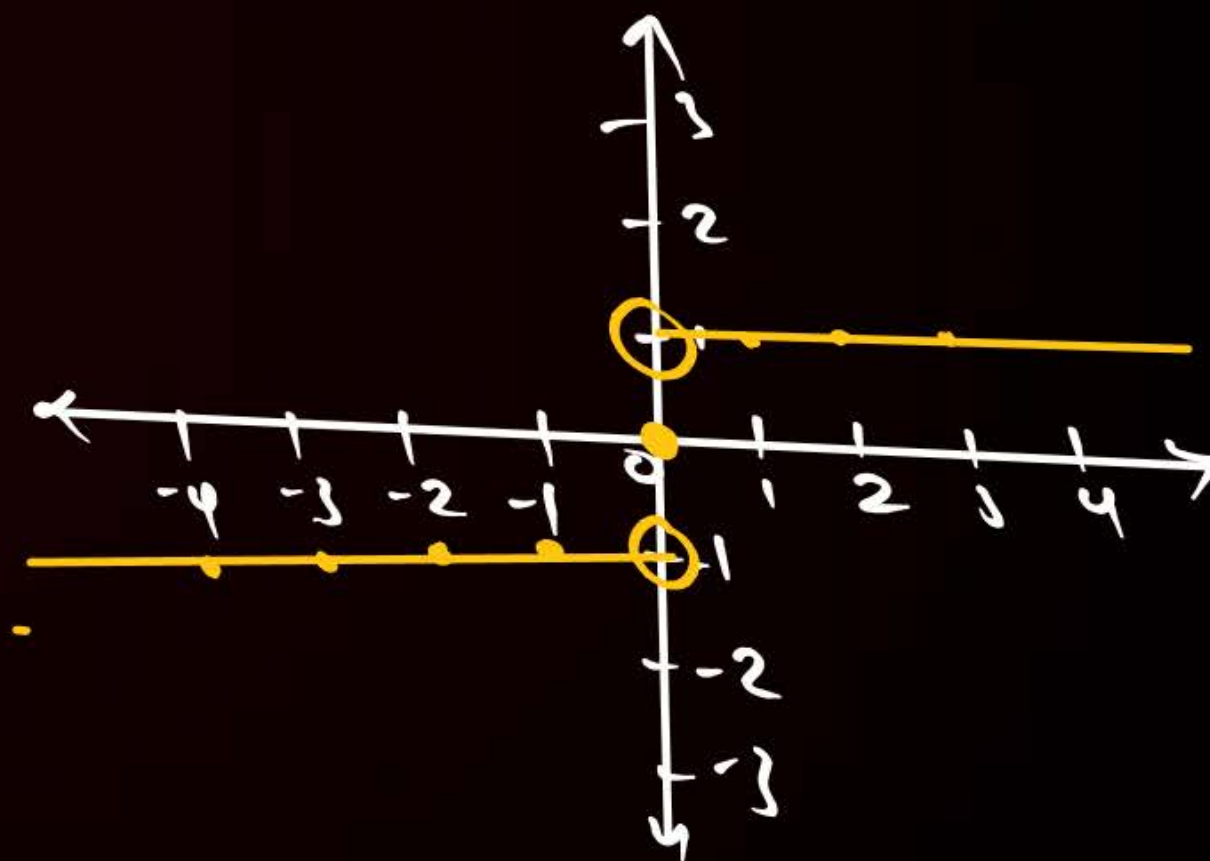
SIGNUM FUNCTION The function f defined by

$$f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} \text{ or } f(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

is called the signum function.

x	0	1	2	3	-1	-2	-3
y	0	1	1	1	-1	-1	-1

Domain: $\mathbb{R}$
Range: $\{1, 0, -1\}$





SOME STANDARD REAL FUNCTIONS AND THEIR GRAPHS



GREATEST INTEGER FUNCTION (FLOOR FUNCTION) For any real number x , we use the symbol $[x]$ or, $\lfloor x \rfloor$ to denote the greatest integer less than or equal to x .

$$[2.2] = 2$$

$$[5.9999] = 5$$

$$[2] = 2$$

$$[-5.5] = -6$$

$$[3] = 3$$

$$[5.5] = 5$$

$$[3.75] = 3$$

$$[-9.8] = -10$$



$$F(x) = [x]$$

$$0 \leq x < 1, \quad F(x) = 0$$

$$1 \leq x < 2, \quad F(x) = 1$$

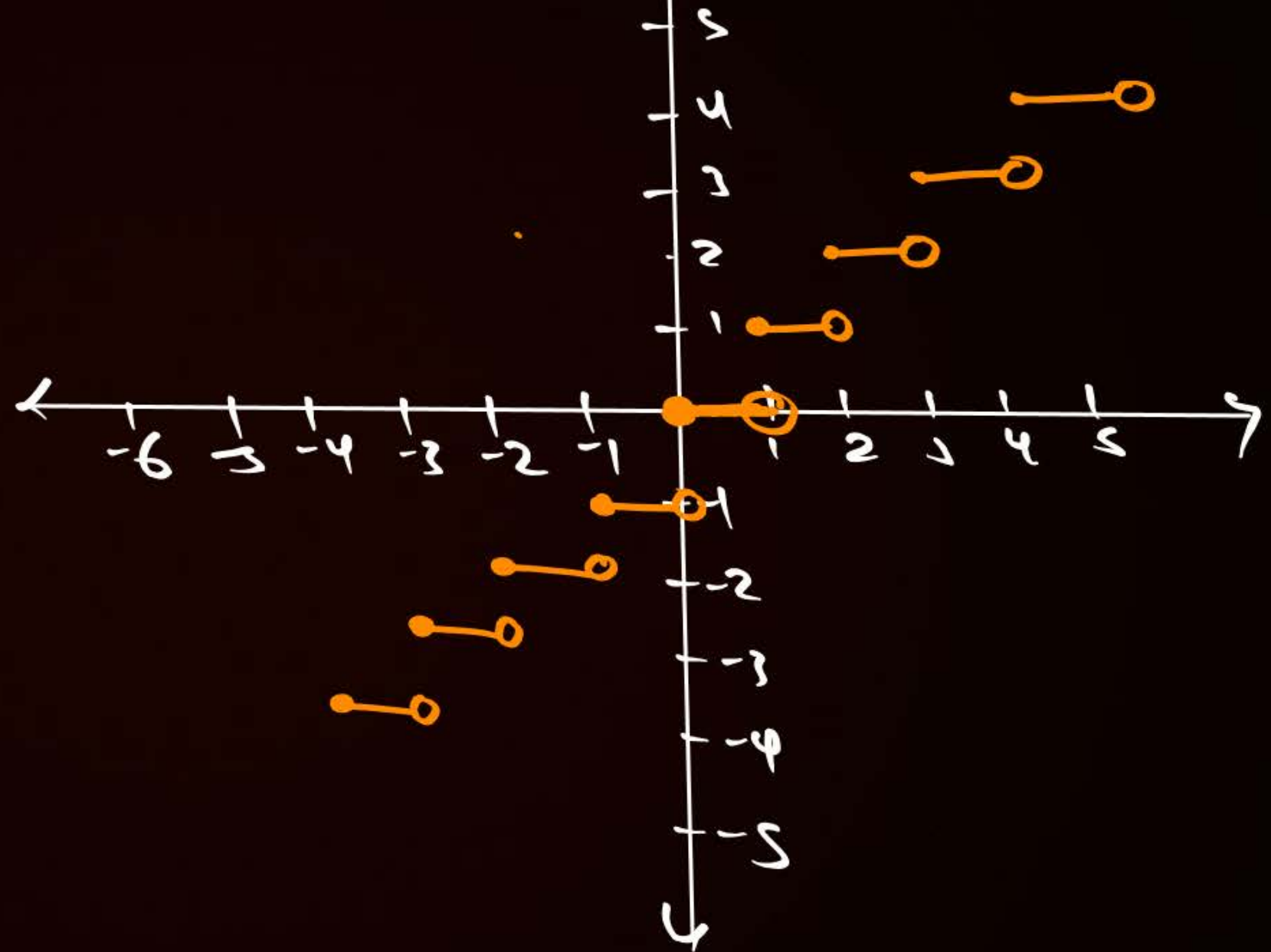
$$2 \leq x < 3, \quad F(x) = 2$$

$$3 \leq x < 4, \quad F(x) = 3$$

$$-1 \leq x < 0, \quad F(x) = -1$$

$$-2 \leq x < -1, \quad F(x) = -2$$

Floor Function / Step Function



A stylized illustration of a person standing on a dark hill, looking out over a landscape of rolling hills under a large, bright sun. The scene is rendered in warm orange and red tones.

THANK YOU